

RISHIKESH S

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SUMMARY

Full-stack engineer with **4+ years** shipping production systems across the stack. Founding engineer at Mando, an AI platform for Workday customers, taking **five product surfaces** from first commit to its **first paid tier** in under two years as sole engineer. Previously built low-latency trading systems at Quantitative Brokers.

EXPERIENCE

Mando

Remote

Founding Software Engineer

Sep 2024 – Present

- Architected the stack as sole engineer: **Next.js** and **FastAPI** on **PostgreSQL**, all environments in **Terraform**.
- Engineered **conversational search** over Workday's knowledge base, owning retrieval, ranking and cited answer generation (**RAG**), so users get a verifiable answer instead of a list of documents.
- Shipped an **expert consultation product** end to end: time-zone-aware booking, live video and recording on Azure Communication Services, and a browser-based editor for trimming and annotating footage.
- Built the **service-management platform** behind it (ticket intake, routing, expert assignment, resolution) and a **Chrome extension** that records a workflow once and generates step-by-step documentation.
- Launched Mando's **first paid tier** by integrating Stripe subscriptions, checkout, invoicing and entitlements.

Quantitative Brokers

Chennai, India

Software Engineer

Jul 2022 – Sep 2024

- Designed a **real-time order monitoring system** processing **200 messages/sec** from multiple upstream sources into live aggregates and market data ladders for trading analysts.
- Cut risk control load time by **over 97%** (40 s to under 1 s) by moving data loading server-side and paginating.
- Reduced critical-path latency by **20%** and CPU usage by **50%** in a C# WPF order entry app on Bloomberg data.
- Implemented **DV01** risk calculations for cash instruments, enabling the desk to expand its cash trading stream.
- Developed capture-and-replay tooling that reproduces production trading incidents from recorded message flow.

Software Development Intern

May 2021 – Jul 2021

- Prototyped **OpenFin** interop across four trading apps and an Angular curve plotter over **kdb+**; both shipped.

Prescribe

Remote

Android Developer

Feb 2021 – May 2021

- Developed a **Firestore**-backed Android app for patient intake and feedback, with a custom notification system and analytics on every step of the flow.

Workfence Technologies

Remote

Software Development Intern

May 2020 – Jul 2020

- Built a **geofencing attendance system** and data visualisation interface designed to scale from 1,000 to **100,000 users**, plus a social-distancing Android app paired with distance-measuring hardware.

PROJECTS

leak | *Go, Bubble Tea, SQLite, GitHub Actions*

github.com/RishikeshSreekumar/leak

- Built a **local-first terminal app** that tracks and audits recurring subscriptions, shipped as a **single static Go binary** for **six OS/arch targets** through a Homebrew tap and CI releases.

deadfall | *TypeScript, Node.js, ts-morph*

npmjs.com/package/deadfall

- Published an **npm CLI** that maps a Next.js app's component graph and flags components nothing renders.
- Exits non-zero when dead components are found, so it doubles as a **CI gate**, and generates an interactive HTML report for exploring the import graph.

mailcleaner | *Python, Textual, SQLite, IMAP, OAuth*

github.com/RishikeshSreekumar/gmail_cleaner

- Developed a **metadata-only** mailbox index across Gmail, Outlook, Fastmail, iCloud and any IMAP server, with terminal and browser UIs for bulk archive, trash and label actions.
- Designed a **no-permanent-delete** safety model: every bulk action previews a sample first, and archive and trash are refused on servers without IMAP MOVE rather than emulated.

Bansuri | *SvelteKit, TypeScript, Web Audio API*

flute.rishikeshs.dev

- Created a flute learning app with **57 lessons** across 11 units and 8 practice tools, including **real-time pitch and drift detection** with all audio processed in the browser.
- Prerendered every lesson as a linkable, indexable URL, each opening its practice tool as a preset link.

Rubik's Cube Solver | *React, Three.js, Web Workers, PWA*

- Implemented **Kociemba's two-phase algorithm** in a Web Worker, with webcam sticker scanning and an animated, orbitable 3D solution that works offline as a PWA.
- Added cube-state validation that names why a cube is impossible: twisted corner, flipped edge or parity.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, C#, Java, SQL

Frontend: React, Next.js, Angular, SvelteKit, Astro, Tailwind CSS, Three.js, WPF

Backend: FastAPI, SQLAlchemy, Node.js, NestJS, .NET, REST APIs, WebSockets, Stripe

Applied AI: LLMs, Retrieval-Augmented Generation (RAG), pgvector, Model Context Protocol (MCP)

Databases: PostgreSQL, Redis, SQLite, kdb+

Cloud & DevOps: Amazon Web Services (AWS), Azure, Terraform, Docker, GitHub Actions, CI/CD, Vercel

Tools: Git, Playwright, Alembic, Chrome Extensions, Web Workers, shell scripting

EDUCATION

Indian Institute of Technology Madras

Chennai, India

Bachelor of Technology (B.Tech.), Chemical Engineering | CGPA: 8.58/10

2018 – 2022

- **All India Rank 2,275** in JEE Advanced 2018; **State Rank 5** in KEAM 2018 (Kerala engineering entrance).